الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني

الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام ٢٠٢٢ الاولى

الاسم والشهرة: ----------
رقم المرشح: ----------

الاختصاص والرمز : الالكترونيك (٢٠٢٠٣)
المادة : الالكترونيك الرقمي
المستندات المسموح بها : آلة حاسبة غير مبرمجة

المعدل : (١٨)
المدة : ساعتان

الإجابات على "كراس الإجابة"

A- 6809

1- ما محتوى الذاكرة $0020 في نهاية كل من البرامج التالية ؟ (6 pts)

a-
```
LDA#$3C
DECA
LSRA
STA$0020
END
```

b-
```
LDB#$03
LDA#$04
MUL
STB$0020
END
```

c-
```
LDA#$B1
SUBA#$02
ANDA#$AA
STA$0020
END
```

d-
```
LDA#$33
ADDA#$1F
STA$0020
END
```

e-
```
LDD#$3E5C
ASRA
STD$0020
END
```

f-
```
LDX#$0020
LDA#$FF
STA, X+
END
```

2- اكتب البرنامج لحل المعادلة S = M + 2N -3 حيث

Write the program that resolves the equation : S = M + 2N -3 where S is the memory $0020, M the memory $0021 and N the memory $0022. **(2 pts)**

3- **Write the program to down count from 9 to 0 on a common cathode 7 segment display connected to PORTB of the PIA through an appropriate BCD-7segment decoder.** **(2 pts)**

B- PIC18F84A

4- **What is the content of the memory 0X30 at the end of each of the following programs?** **(6 pts)**

a-
```
MOVLW 0X3C
BCF STATUS,C
MOVWF 0X30
RRF 0X30,1
END
```

b-
```
MOVLW 0XBB
MOVWF 0X30
MOVLW 0XEE
ANDWF 0X30,1
END
```

c-
```
MOVLW 0XBB
MOVWF 0X30
MOVLW 0XEE
IORWF 0X30,1
END
```

d-
```
MOVLW 0XC1
MOVWF 0X20
BSF STATUS,C
RLF 0X20,0
MOVWF 0X30
END
```

e-
```
MOVLW 0X95
ADDWF 0X37
MOVWF 0X30
DECF 0X30,1
END
```

f-
```
MOVLW 0XBB
MOVWF 0X30
MOVLW 0XAA
SUBWF 0X30,1
END
```

5- **Choose the correct answer:** **(4 pts)**

1- **To turn OFF a LED whose cathode is connected to RA5 and anode to Vcc, we write:**
a- BCF PORTA,5 b- BSF PORTA,5 c- BSF TRISA,5

2- **To turn ON 8 LED whose cathodes are connected to PORTB, we write:**
a- BSF PORTB b- CLRF PORTB c- MOVLW 0XFF
MOVWF PORTB

3- **Having W= (29)H and the memory 0X20 that contains (B9)H, if we execute « ADDWF 0X20,F », we obtain:**
a- W= 0XFA b- W= 0XEF c- 0X20= 0XE2

4- **Having W= (10)H and the memory 0X20 that contains (B9)H, if we execute « SUBWF 0X20,F », we obtain:**
a- W= 0XA9 b- W= 0X10 c- 0X20= 0X10

ANNEXE / APPENDIX

Instruction set of 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

PIA:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**

Instruction set of PIC 16F84A

ADDWF, ANDWF, CLRF, CLRW, COMF, DECF, DECFSZ, INCF, INCFSZ, IORWF, MOVF, MOVWF, NOP, RLF, RRF, SUDWF, SWAPF, XORWF, BCF, BSF, BTFSC, BTFSS, ADDLW, ANDLW, CALL, CLRWDT, GOTO, IORLW, MOVLW, RETFIE, RETLW, RETURN, SLEEP, SUBLW, XORLW.

Internal architecture of PIC16F84A

File Address			File Address
00h	Indirect addr.(1)	Indirect addr.(1)	80h
01h	TMR0	OPTION_REG	81h
02h	PCL	PCL	82h
03h	STATUS	STATUS	83h
04h	FSR	FSR	84h
05h	PORTA	TRISA	85h
06h	PORTB	TRISB	86h
07h	—	—	87h
08h	EEDATA	EECON1	88h
09h	EEADR	EECON2(1)	89h
0Ah	PCLATH	PCLATH	8Ah
0Bh	INTCON	INTCON	8Bh
0Ch	General Purpose Registers (SRAM)	Mapped (accesses) in Bank 0	8Ch
			FFh
	Bank 0	Bank 1	

PS2:PS0: Prescaler Rate Select bits

Bit Value	TMR0 Rate	WDT Rate
000	1 : 2	1 : 1
001	1 : 4	1 : 2
010	1 : 8	1 : 4
011	1 : 16	1 : 8
100	1 : 32	1 : 16
101	1 : 64	1 : 32
110	1 : 128	1 : 64
111	1 : 256	1 : 128